

Module 05

Docker Compose



Docker Compose

DevOps end to end services & solutions

Agenda

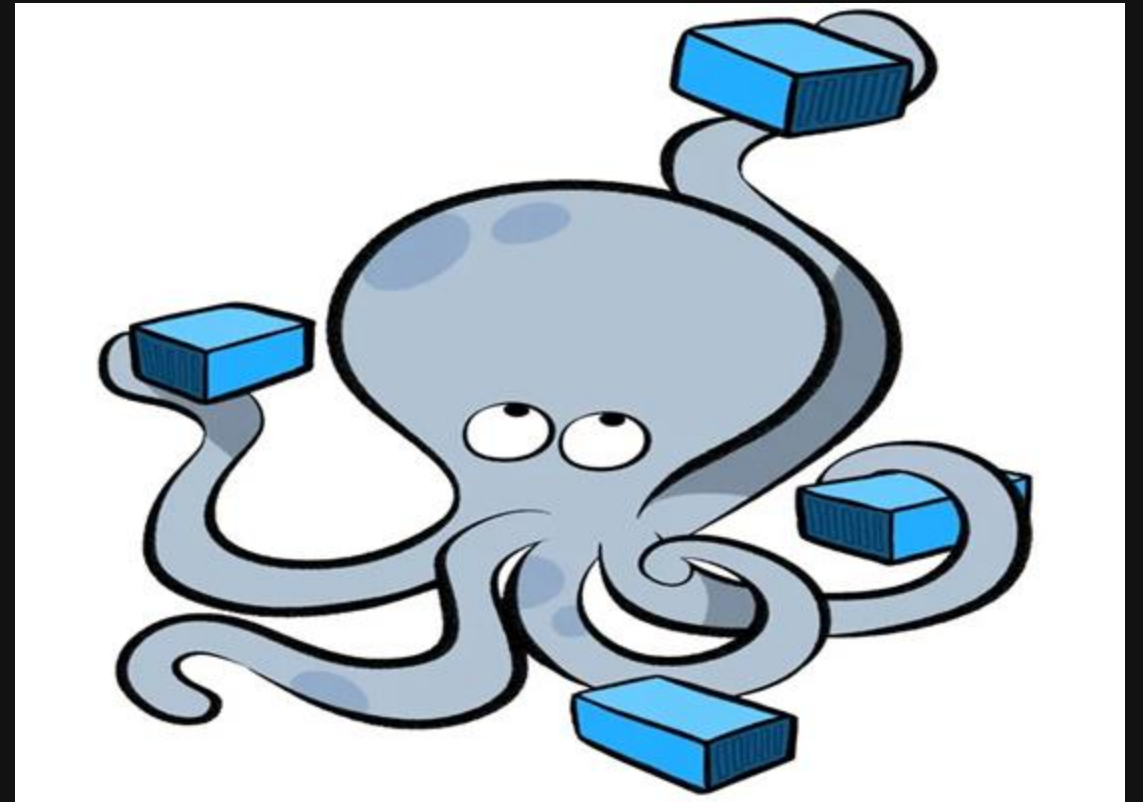
- > Docker-Compose Overview
- > Docker-Compose Commands
- > YAML Configuration (docker-compose.yml)
- > Key Features & Usage
- > Hands-On – Lab-05 – Docker-Compose

Docker-Compose Overview

What is it?

A tool to define and manage multi-container Docker applications using a single YAML file.

- > Run multiple services with one command.
- > Simplifies control of the entire application stack.
- > Manages Services, Networks, and Volumes.



Basic Commands



up -d

Start services in the background.



down

Stop and remove containers & networks.



build

Build images before starting services.



ps

List running services defined in the YAML.

YAML Configuration

A sample docker-compose.yml file:

```
version: "3.8"
services:
  flask-app:
    build: .
    container_name: flask_app
    ports:
      - "5000:5000"
    environment:
      - MONGO_URI=mongodb://mongo:27017/mydb
    depends_on:
      - mongo
  mongo:
    image: mongo:latest
    container_name: mongo_db
```

Key Elements

- > **version:** Compose file format version.
- > **services:** The containers to run.
- > **depends_on:** Start order control.
- > **volumes:** Persistent storage.

Key Features & Usage

★ Features

- **Multi-container Deployment:** All apps in one file.
- **Service Orchestration:** Automate dependencies.
- **Simplified Networking:** Hostnames = Service Names.
- **Volume Management:** Persistent data.
- **Environment Config:** Supports .env files.

🚀 Common Usage

- Local development with complex stacks.
- Running microservices architecture locally.
- Testing in CI/CD pipelines.
- Setting up reproducible environments quickly.

Hands-On – Lab 05

Docker Compose Application

Create an application that reads and writes from a database.

- > **DB Options:** MSSQL, Mongo, Queue, JSON, etc.
- > **Requirement:** All services must be in `docker-compose.yml`.
- > **Task:** Login to UI, input text, and verify data in the DB.

Questions?

Thank you for participating.

Feel free to reach out:
<https://dinghy-e2e.com>

